

Project Panopticon

Zero-Escalation Support through Adversarial Multi-Agent Orchestration

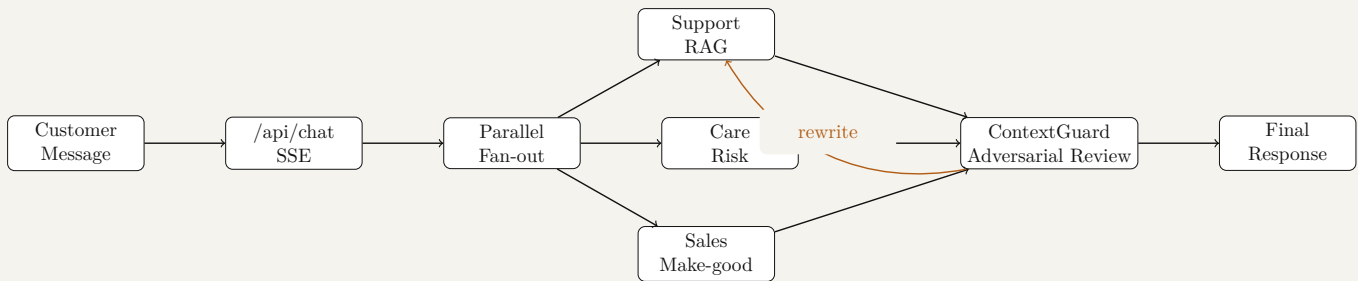
Hackathon Gold-Tier Submission

Core idea: A customer escalation should not trigger a chatbot. It should trigger a private war room. Panopticon runs Support, Customer Care, and Sales agents in parallel, then lets ContextGuard attack their drafts before a single customer-facing word is released.

Why It Matters

Enterprise escalations combine technical failure, churn risk, and commercial opportunity. Traditional bots flatten those into one generic answer. Panopticon separates the concerns, lets specialized agents reason independently, and uses an adversarial adjudicator to prevent hallucinations, policy violations, and tone mistakes.

Architecture



Implemented Features

- Split-screen Next.js interface with customer chat on the left and a streaming Live War Room terminal on the right.
- Local vector database built from a synthetic enterprise knowledge base.
- Parallel Support, Care, and Sales agents with strict TypeScript schemas.
- ContextGuard adversarial review with forced rewrite when a deprecated API key appears.
- Churn scoring and automatic 20% SLA credit guardrail.
- Contextual Enterprise make-good offer when the technical problem maps to active-active failover.
- Sales suppression for Enterprise customers and unsupported/trick-question flows.
- Server-Sent Events streaming for logs, drafts, guardrail decisions, and final output.
- OpenAPI documentation at `/api/openapi` plus local architecture and prompt docs.
- Remotion composition for programmatic product video generation.
- Automated tests for retrieval, churn scoring, rewrite behavior, and demo cases.

Evaluation Alignment

Innovation: Panopticon is an adversarial operating loop, not a single chatbot prompt.

Technical Architecture: DAG orchestration, SSE streaming, local RAG, caching, strict schemas, and provider integration points.

Support: Grounded fix generation with self-correction.

Customer Care: Predictive churn scoring and bounded SLA credit.

Sales: Crisis-aware make-good upsell only when useful and appropriate.

Documentation: README, architecture diagram, prompts, OpenAPI, screenshots, demo script, and this submission package.

Demo Moment

The strongest demo prompt is: *“Your API crashed again and my payment gateway is down. I’m taking my business elsewhere!”* The Live War Room shows Support drafting a risky deprecated key instruction, ContextGuard blocking it, Support rewriting the fix, and the customer receiving a safe answer with `gateway_failover=true`, `Idempotency-Key`, and `/v3/status/incidents`.